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Metal Matters

Aluminium - The most bullish set-up in over~50 years, next stop \$4,000/t

CITI'S TAKE

The aluminium market has just been hit by one of the largest supply shocks in its modern history (certainly since the 1970s, when the 1st and 2nd energy shocks led to substantial output cuts). At the same time, spare capacity is near-zero, inventories pre-shock were already at all-time (55-year) lows, and the cost of substitutes such as copper and plastics are extremely high by historical standards. We see aluminium inventories drawing to fresh all-time lows over the next 6-12 months, with the associated buying of physical inventory futures hedges driving up prices to average \$4,000/t during 2H'26, and in our bull case to average \$5,350/t in 2027 (N.B. we've been bullish aluminium from \$2,600/t since mid-2025). We have two open trade recommendations; see details below.

The aluminium market no longer requires strong demand growth to remain structurally tight. Following the Middle East disruption, supply elasticity has fallen across the system as China remains capped and ex-China supply growth is insufficient to offset ongoing deficits. Unlike previous downturns where weaker demand and eventual supply rationalisation loosened the market, the current shock is supply-driven and much of the damage is already done. We expect ~2.7Mt of deficit this year even under sluggish demand. Only a severe recession comparable to the Volcker-era downturn or the 2008/9 GFC appears sufficient to stabilise inventory cover, in our view.

Importantly, persistent deficits cannot remain theoretical. If consumption exceeds production, the imbalance must be absorbed through inventory drawdowns. This is now the core issue facing the market. Initially, inventories can absorb deficits through hidden, financing, merchant and pipeline stocks. As a result, tightening cycles are often volatile, frustrating, and macro-driven rather than immediately explosive. Over time, inventory draw alters market structure as buffers shrink and embedded short hedges tied to financing and carry unwind. Such tightening does not require sustained speculative long accumulation. As buffers shrink and hedges unwind, the market becomes more exposed even without new long positioning. Under these conditions, relatively small shortages can generate disproportionately nonlinear price responses.

This is why aluminium's upside convexity remains strong despite macro uncertainty. While near-term volatility and position reduction may pressure prices, downside appears increasingly self-limiting outside a severe recession. Reflecting the rising probability of inventory-driven tightening, we see a credible path to \$4,000/t aluminium over the next 0-3 months if demand destruction remains insufficient to offset constrained supply. Full forecast [here](#). We have two open trade recommendations – [long Dec26 \[3463 \(5pm BST 18-May-26\)\]](#) and [long 3,300/3,600 call spreads \[132 \(5pm BST 18-May-26\)\]](#), which we opened in January. The main downside risks are a major risk off and/or a major global recession. Note: Futures trading involves substantial risk of loss. Trade ideas owned by **Max Layton**.

[Electricity in the form of metal - sleepwalking into the biggest deficits in over 20 years](#)

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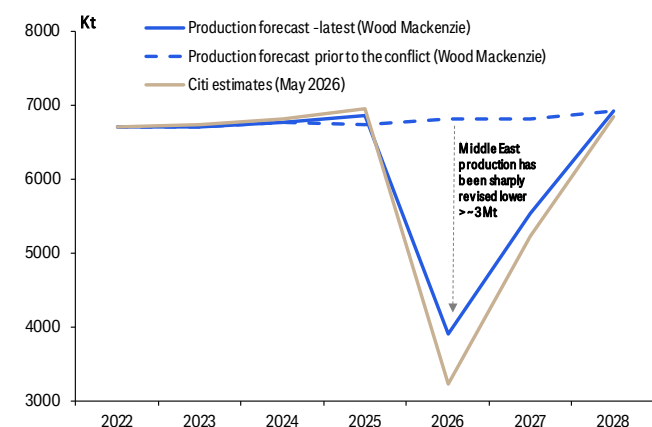
Supply disruptions have shifted aluminium into a structurally tighter regime

The aluminium market is increasingly transitioning from an initial geopolitical shock into a structurally tighter inventory regime. The first phase of the move was dominated by war escalation risk, Strait of Hormuz concerns, oil price spikes, and broader macro uncertainty. Two months later, however, the more important question is whether the global aluminium system remains structurally undersupplied even if the Middle East disruption does not materially worsen from here.

The answer increasingly appears to be yes. Prior to the US-Iran conflict, consensus expectations broadly assumed relatively stable GCC aluminium production over the coming years. That assumption no longer holds. More than 3Mt of production losses are now embedded into revised production forecast, while the recovery path itself remains highly uncertain and dependent on conflict duration, infrastructure repair timelines, logistics normalization, and raw material restocking. In practice, the market is unlikely to experience a rapid V-shaped recovery in regional supply.

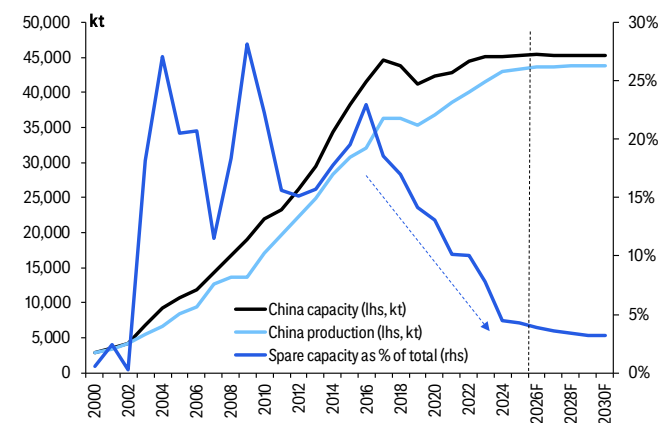
Supply elasticity has also fallen materially across the broader aluminium system. China now operates under effective capacity ceilings following years of supply-side reform, while much of the currently profitable global capacity is already running. Although Indonesia remains one of the few areas capable of delivering meaningful ex-China growth, the pace and timing of future expansion still carry execution and ramp-up risks. As a result, even relatively sluggish demand growth increasingly appears sufficient to generate persistent deficits.

Figure 1. Middle East–latest forecast reflects currently assessed disruption; 2027–2028 recovery remains subject to conflict duration, infrastructure damage, and restart constraints



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Source: Citi Research, Wood Mackenzie

Figure 2. Historical spare capacity in China's aluminium industry has disappeared—implications for prices and supply elasticity



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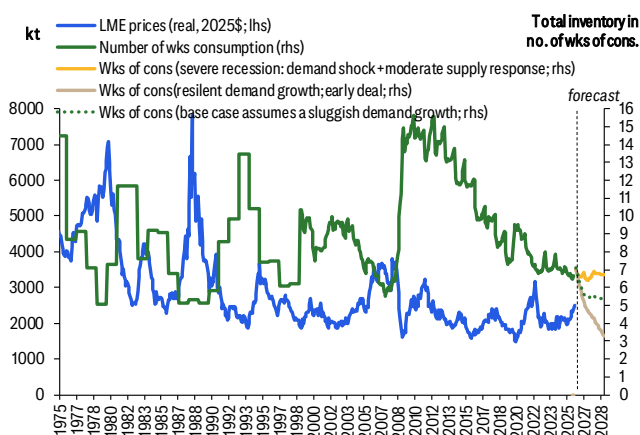
Severe recession now the only viable path to inventory stability

Only an extreme recession scenario comparable to the Volcker-era downturn or the 2008–09 GFC appears sufficient to stabilise aluminium inventory cover. We have already revised down our aluminium demand growth assumptions materially in our base case supply and demand model (see Figure 11), yet even under this more sluggish framework, the market continues drawing inventories lower over time.

Our severe recession scenario assumes a moderate primary aluminium demand decline in 2026 followed by a much sharper contraction in 2027 (broadly mirroring previous major crisis periods; more than -5% y/y), alongside a further decline into 2028. Even under those assumptions, inventory cover only broadly stabilises rather than rebuilding meaningfully. Importantly, previous aluminium downturns were typically eventually loosened not only by weaker demand, but also by delayed supply rationalisation as lower prices and margin compression forced production cuts over time. By contrast, the aluminium market today is already dealing with a large supply shock following the Middle East disruption. As a result, even severe recession-level demand destruction now appears insufficient to generate the kind of meaningful inventory rebuilding seen in previous downturns.

Historically, severe recessions rebuilt aluminium inventories. In the current framework, they merely prevent inventories tightening further. This represents a materially higher threshold for market rebalancing relative to previous aluminium cycles. The market now requires not simply weaker demand, but outright severe recession conditions combined with additional supply adjustment merely to stabilise inventory cover.

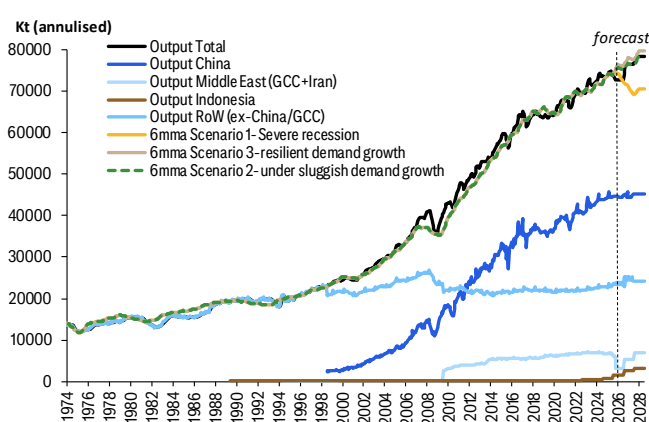
Figure 3. Historically severe recessions rebuilt inventories. Today, they merely stabilise inventory cover



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Source: Citi Research, CRU, Wood Mackenzie

Figure 4. With ME supply disrupted, Indonesia remains the only viable ex-China growth option, but at limited scale



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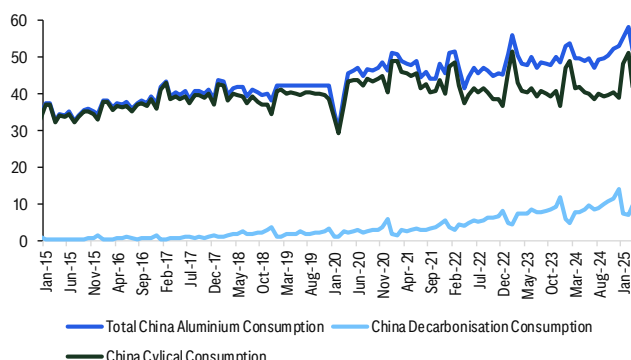
Source: Citi Research, Wood Mackenzie

Today's aluminium demand profile also looks structurally less cyclical than during previous recession periods. Historically, aluminium demand growth relied much more heavily on traditional industrial cyclicality. By contrast, a much larger share of current demand growth is linked to power grids, renewable energy infrastructure and electrification supply chains.

This distinction matters because energy transition demand tends to be more policy-supported and less sensitive to cyclical slowdowns than traditional

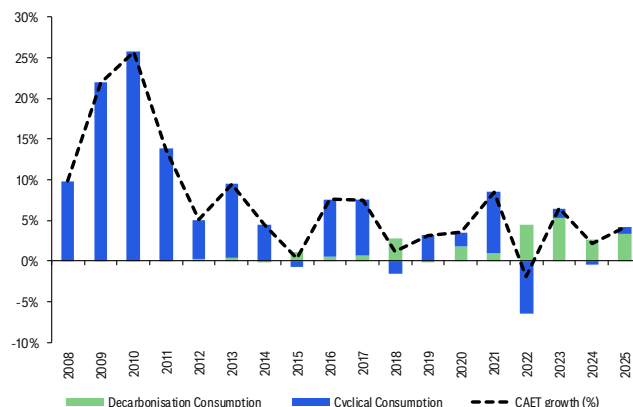
industrial demand. In our end use demand tracker (CAET), energy transition demand now represents close to a quarter of Chinese aluminium demand. As a result, aluminium demand today likely exhibits lower downside elasticity than during previous recession episodes.

Figure 5. China's demand mix increasingly reflects a broader structural shift in aluminium consumption



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Source: Citi Research, NBS, SMM, Bloomberg

Figure 6. China's aluminium demand has become increasingly supported by less cyclical energy transition sectors



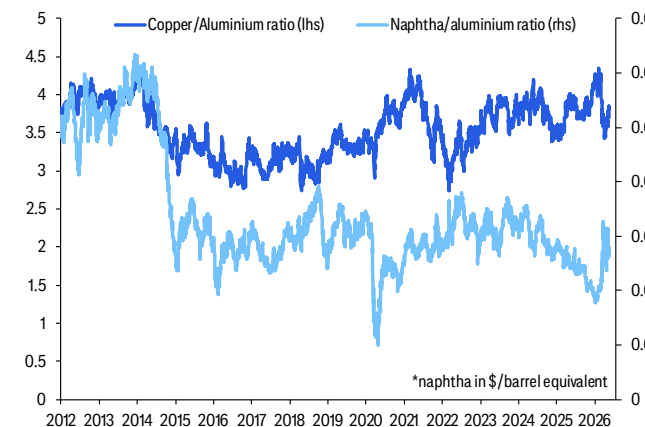
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We also expect the impact of the current US-Iran conflict on aluminium demand to remain highly asymmetric between China and ex-China markets. A similar pattern was visible during the 2008-09 GFC when ex-China economies absorbed most of the cyclical demand destruction while Chinese demand proved comparatively resilient. The difference today is that China now accounts for a substantially larger share of global aluminium consumption. Based on our adjusted end-use estimates, China demand is approaching 60% of the global total.

The market's traditional self-correcting mechanisms also appear materially weaker than in previous aluminium cycles. During earlier periods of elevated prices, China retained the ability to restart or add sizeable volumes of supply relatively quickly. That mechanism has weakened materially following years of supply-side reform and effective capacity ceilings.

Higher aluminium prices may still prove less effective at triggering broad substitution away from aluminium than during previous cycles. Copper-to-aluminium ratios remain historically elevated (Figure 7), continuing to support substitution towards aluminium across multiple end-use sectors, while petrochemical feedstock costs also remain structurally higher versus the early-2010s. Relative aluminium-to-steel pricing (Figure 8) has become increasingly elevated in China, although substitution economics across ex-China markets remain materially less extreme. In practical terms, downstream consumers increasingly face a world where competing material systems are also expensive, limiting the immediacy of broad-based substitution away from aluminium.

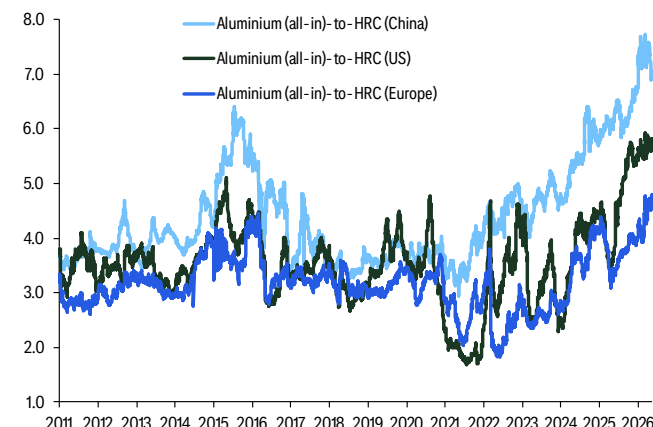
Figure 7. Aluminium remains cheap-to-average versus copper and plastic (naphtha as the proxy)



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Source: Citi Research, Bloomberg

Figure 8. Relative aluminium-to-steel pricing has become increasingly elevated in China, although substitution economics across ex-China markets remain materially less extreme



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Persistent deficits cannot remain theoretical accounting concepts. If consumption continues exceeding production, the imbalance must ultimately be absorbed through inventory drawdowns. This is increasingly the core issue facing the aluminium market.

Initially, inventories can quietly absorb deficits for a period of time. Hidden inventory, financing inventory, merchant stock and pipeline inventory can all continue absorbing shortages quietly without immediately generating aggressive price responses. This helps explain why commodity tightening cycles are often initially volatile, frustrating, and macro-dominated rather than immediately explosive.

Over time, however, inventory draw begins altering market structure itself. Aluminium inventories historically represented not only physical buffers but also a significant source of embedded futures short hedges through financing and carry structures. As inventories decline, physical holders simultaneously reduce associated short hedges, gradually shrinking the market's embedded short base.

This is why later-stage commodity squeezes often become increasingly nonlinear. Once inventories cease functioning as effective buffers, nearby tightness intensifies, backwardation structures steepen, and physical optionality disappears. Under those conditions, relatively small additional shortages can begin generating disproportionately large price responses.

The key risk for the aluminium market is therefore not necessarily demand weakness itself, but insufficient demand destruction relative to structurally constrained supply. Aluminium no longer requires exceptionally strong global growth to tighten materially further. It merely requires demand to avoid collapsing.

The largest upside risk emerges if the energy shock ultimately proves less damaging to economic activity than markets currently fear. Under that scenario, persistent deficits and ongoing inventory depletion would likely tighten market structure progressively over time, increasing the probability of a nonlinear upside price response later in the tightening cycle.

While near-term aluminium price action may remain volatile and macro-dominated, we increasingly believe downside becomes self-limiting outside a severe recession scenario. Macro-driven de-risking, weaker cross-asset sentiment and positioning reductions may still generate sharp short-term corrections. However, unless demand destruction becomes large enough to offset the current supply losses, inventory tightening is likely to persist over time.

Historically, however, commodity squeezes rarely begin with immediate straight-line rallies. Tightness initially emerges quietly through inventory depletion before progressively reshaping market structure itself. As inventories decline, the market loses not only physical buffers but also embedded short hedges associated with financing and carry structures.

Reflecting the increasing probability of such an inventory-driven tightening regime, we now see a credible path towards \$4,000/t aluminium over the next 0-3 months should demand destruction remain insufficient to offset structurally constrained supply. Under those conditions, aluminium's price elasticity could rise materially as nearby tightness intensifies and the market becomes increasingly sensitive to marginal shortages.

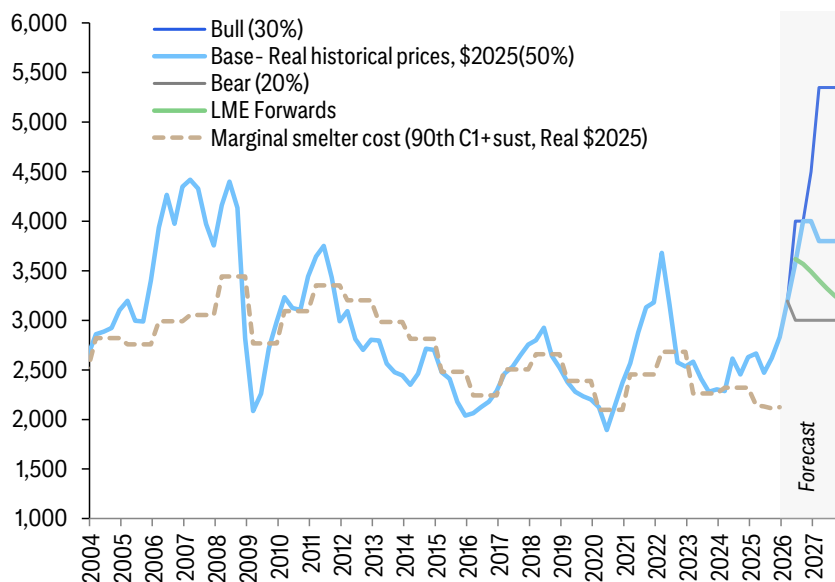
Figure 9. Citi aluminium price forecasts and scenarios

Price Forecasts	Scenario Weight	Unit	0-3 mth pt price	6-12 mth pt price	1Q26	2Q26f	3Q26f	4Q26f	2026f	1Q27f	2Q27f	3Q27f	4Q27f	2027f
Aluminium (Bull)	30%	\$/t				4000	4000	4500	3925	5350	5350	5350	5350	5350
Aluminium (Base)	50%	\$/t	4000	4000	3195	3580	4000	4000	3700	3800	3800	3800	3800	3800
Aluminium (Bear)	20%	\$/t				3000	3000	3000	3050	3000	3000	3000	3000	3000

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Figure 10. Citi aluminium price forecasts and scenarios



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Figure 11. Aluminium supply and demand balance

kt	2020	2021	2022	2023	2024	2025	2026F	2027F	2028F
Total Aluminium Consumption	87136	95003	96937	99789	103133	106362	108336	110741	115606
<i>y/y chg</i>	-3.6%	9.0%	2.0%	2.9%	3.4%	3.1%	1.9%	2.2%	4.4%
Recycled Metal Consumption	26,192	27,918	30,402	31,345	32,224	33,795	35,012	37,663	39,297
<i>y/y chg</i>	-6.6%	6.6%	8.9%	3.1%	2.8%	4.9%	3.6%	7.6%	4.3%
Primary Consumption	62,618	68,816	68,577	70,003	72,660	74,192	75,499	76,184	78,137
Consumption Incr. (%)	-2.8%	9.9%	-0.3%	2.1%	3.8%	2.1%	1.8%	0.9%	2.6%
<i>Of Which China</i>	<i>37,466</i>	<i>40,047</i>	<i>40,183</i>	<i>42,692</i>	<i>44,762</i>	<i>45,960</i>	<i>47,090</i>	<i>47,624</i>	<i>49,093</i>
<i>Consumption Incr (%)</i>	<i>4.0%</i>	<i>6.9%</i>	<i>0.3%</i>	<i>6.2%</i>	<i>4.8%</i>	<i>2.7%</i>	<i>2.5%</i>	<i>1.1%</i>	<i>3.1%</i>
Primary Production	64,751	67,402	68,843	70,691	72,713	74,165	72,729	76,501	78,306
Supply Incr (%)	2.3%	4.1%	2.1%	2.7%	2.9%	2.0%	-1.9%	5.2%	2.4%
<i>Of Which China</i>	<i>36,745</i>	<i>38,524</i>	<i>40,085</i>	<i>41,586</i>	<i>43,071</i>	<i>43,877</i>	<i>44,964</i>	<i>45,121</i>	<i>45,215</i>
<i>Supply Incr (%)</i>	<i>3.9%</i>	<i>4.8%</i>	<i>4.1%</i>	<i>3.7%</i>	<i>3.6%</i>	<i>1.9%</i>	<i>2.5%</i>	<i>0.3%</i>	<i>0.2%</i>
<i>Of Which Indonesia</i>	<i>245</i>	<i>243</i>	<i>229</i>	<i>372</i>	<i>533</i>	<i>806</i>	<i>1,530</i>	<i>2,666</i>	<i>3,130</i>
<i>Supply Incr (%)</i>	<i>-2%</i>	<i>-1%</i>	<i>-6%</i>	<i>63%</i>	<i>43%</i>	<i>51%</i>	<i>90%</i>	<i>74%</i>	<i>17%</i>
Smelter Capacity (kt/y)	75,132	76,237	78,553	79,026	79,260	79,747	81,449	82,505	83,427
Primary Surplus/Deficit (Base Case)	2216	-1384	366	750	179	-27	-2770	317	169
<i>China</i>	<i>339</i>	<i>81</i>	<i>392</i>	<i>307</i>	<i>345</i>	<i>296</i>	<i>-286</i>	<i>1,069</i>	<i>-835</i>
<i>Ex-China</i>	<i>1,877</i>	<i>-1,465</i>	<i>-26</i>	<i>443</i>	<i>-166</i>	<i>-323</i>	<i>-2,483</i>	<i>-752</i>	<i>1,004</i>
US\$/t	1,705	2,476	2,707	2,255	2,425	2,628	3,700	3,800	3,600

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